HarvestWatch: An AI-Powered Early Warning System for Food Insecurity in Rwanda

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Track: Track 1 — Agricultural Productivity
Team: HarvestWatch
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GitHub Repository: https://github.com/BodeMurairi2/food_supply_earlyWarning
Demo Video: https://youtu.be/Z_OSBGWulKI

Executive Summary

Food insecurity remains an important development challenge in Rwanda. In 2024, approximately 586,000 households in Rwanda, representing 17% of all households, experienced food insecurity. A further 1.5 million households were marginally food secure, meaning that a poor harvest, an increase in food prices or another shock could push them into food insecurity. The situation is not the same across the country. While approximately 4% of households in Kigali City were food insecure, the proportion reached 23% in the Western Province. At district level, the reported share ranged from below 2% in Kicukiro to 38% in Nyamasheke [3]. These differences demonstrate why national averages alone are not sufficient for understanding local food-security conditions. The situation is not the same across the country. While approximately 4% of households in Kigali City were food insecure, the proportion reached 23% in the Western Province. At district level, the reported share ranged from below 2% in Kicukiro to 38% in Nyamasheke [3]. These differences demonstrate why national averages alone are not sufficient for understanding local food-security conditions.

Rwanda currently collects important information about household food security through the Comprehensive Food Security and Vulnerability Analysis (CFSVA), conducted by the National Institute of Statistics of Rwanda (NISR), the Ministry of Agriculture and Animal Resources (MINAGRI) and the World Food Programme (WFP). However, the assessment is conducted approximately every three years. The 2024 survey collected information between April and June 2024, but its final report was published in July 2025 [3]. During the years between assessments, rainfall, agricultural production and food prices continue to change. For example, food prices increased by approximately 60% between April 2021 and May 2024, a period that fell between two CFSVA rounds [3]. Decision-makers therefore need additional information to identify districts that may require closer monitoring before the next household survey becomes available.

HarvestWatch is a working district-level early warning platform developed to help address this information gap, and help stakeholders act early and support districts in need. It combines historical CFSVA information with agricultural statistics from NISR, rainfall estimates from CHIRPS, temperature information from ERA5-Land and food-price data from WFP.

The platform uses a classical machine learning model, Random Forest to estimate the percentage of households experiencing moderate or severe food insecurity in each district. It then converts the numerical estimate into a project-defined Low, Medium or High risk category. The model was evaluated against 60 genuine CARI survey observations from 2021 and 2024. It achieved a mean absolute error of 5.39 percentage points, compared with 7.32 percentage points for a baseline that assumes the previous survey result remains unchanged. This represents approximately 26% lower average error. On the 2024 observations, the model achieved an MAE of 4.44 percentage points, compared with 7.29 for the baseline.

HarvestWatch translates these results into an accessible web application built with React and FastAPI. It provides a national overview, district-level forecasts, historical model

evaluation, agricultural and climate visualizations, a Prediction Lab for testing hypothetical scenarios, SHAP-based explanations and an AI Analyst that retrieves relevant external evidence. The application also includes automated tests to check the reliability of its core services and safeguards.

The 2026 forecasts identify Rutsiro, Ngororero and Nyamasheke as High under the project's risk thresholds, with an average district estimate of 19.6% across Rwanda. These are model-generated estimates, not official survey measurements. Their purpose is to help decision-makers identify where further investigation may be appropriate.

HarvestWatch is designed to support MINAGRI, NISR, district authorities, WFP and development partners by making existing information more connected, accessible and useful. It does not replace official statistics, field assessments or institutional decision-making. Its purpose is to provide an additional evidence-based signal during the years when new representative household food-security measurements are unavailable.

1. Problem Understanding and Relevance

1.1 Agriculture and food security in Rwanda

Agriculture is one of the foundations of Rwanda's economy and an important source of household livelihoods. According to NISR's national accounts, agriculture contributed approximately 25% of Rwanda's GDP in 2024 and recorded growth of about 5% during the year [1]. The sector also remains a major source of employment, accounting for approximately 39.9% of employed people according to the 2024 Labour Force Survey [2].

The importance of agriculture extends beyond its contribution to the economy. Many households depend on agricultural production for both food and income. In Season A 2024, approximately 1.372 million hectares of Rwanda's 2.376 million hectares of land were used for agriculture, representing about 58% of the total land area [4]. The country produced an estimated 507,985 tonnes of maize during that season, approximately 30% more than in Season A 2023. Bean production reached 233,142 tonnes, an increase of approximately 18% compared with the previous year [4].

These figures illustrate how agricultural production can change between seasons. Such changes affect the availability of food, household income and the supply of agricultural products in local markets. When production decreases, households that depend on farming may have less food to consume or sell. When production increases, food availability may improve, although the benefits are not necessarily distributed equally among households.

Food security is also influenced by households' access to markets. Even families that produce some of their own food often purchase additional food items. Consequently, changes in staple-food prices can affect household purchasing power regardless of whether

agricultural production has been favourable. A household may have access to sufficient food during a good harvest but struggle to afford food when market prices increase.

Climate conditions add another layer of vulnerability. Rwanda's agricultural production is strongly influenced by rainfall, and many farming households depend on rain-fed agriculture. Changes in rainfall distribution, unusually dry periods, excessive rainfall and temperature conditions can affect crop development and agricultural productivity.

These relationships explain why agricultural productivity and food security are closely connected. Monitoring production alone does not reveal whether households can access sufficient food, while monitoring household food insecurity without considering agricultural and market conditions may make it difficult to understand emerging pressures.

HarvestWatch brings these dimensions together. By combining agricultural production, climate and market information with historical household food-security outcomes, it investigates how changing conditions relate to food insecurity across Rwanda's districts.

1.2 Rwanda's current food-security situation

The 2024 CFSVA provides the most recent comprehensive national assessment of household food security in Rwanda. It covered approximately 9,000 households across 900 enumeration areas and all 30 districts [3].

Its findings show that 83% of households were food secure or marginally food secure, while 16% were moderately food insecure and 1% were severely food insecure. Approximately 1.5 million households were marginally food secure and therefore vulnerable to shocks that could affect their ability to access sufficient food.

Indicator	**2024 CFSVA finding**
Food secure and marginally food secure households	83%
Moderately food insecure households	16%
Severely food insecure households	1%
Moderately food insecure households	Approximately 551,000

Severely food insecure households	Approximately 35,500
Marginally food secure households	Approximately 1.5 million
Food insecurity in the Western Province	23%
Food insecurity in the Southern Province	16%
Food insecurity in Kigali City	4%
Children under five experiencing stunting	30%

Source: NISR, MINAGRI and WFP, CFSVA Rwanda 2024 [3].

The national situation improved between 2021 and 2024. The proportion of households classified as food secure or marginally food secure increased from 79% to 83%, while moderate food insecurity decreased from 19% to 16% and severe food insecurity decreased from 2% to 1% [3].

However, national improvement does not mean that every district experienced the same progress. Food insecurity increased in some locations while declining in others. The Southern Province, for example, experienced an increase from approximately 20.5% in 2018 to 23.4% in 2024 [3].

The team's analysis of district survey results also illustrates these differences. Between the 2021 and 2024 observations, food insecurity decreased in 19 districts and increased in 11. Nine districts changed their project-defined risk category, and the largest district-level change was approximately 23 percentage points.

These differences demonstrate the importance of local monitoring. A national figure can show that overall conditions are improving while concealing districts where households are experiencing worsening food-security conditions. Identifying these differences is important for agricultural planning, local preparedness and the prioritisation of further assessments.

1.3 The information gap between CFSVA surveys

The CFSVA is a major source of representative household food-security information in Rwanda. Historical rounds were conducted in 2006, 2009, 2012, 2015, 2018, 2021 and 2024 [3].

The assessment's scale is important to its reliability. The 2024 round covered 9,000 households in 900 enumeration areas and included household, community and mother-and-child questionnaires. However, conducting an assessment of this size every year would require substantial resources. Consequently, the survey follows an approximately three-year cycle.

This creates a structural information gap.

Year	**2021**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**
CFSVA information	Survey	No survey	No survey	Survey	No survey	No survey	Expected survey

Two out of every three years between consecutive survey rounds have no new CFSVA measurement. During these periods, agricultural production, rainfall and food prices continue to change, but decision-makers do not automatically receive a new representative household food-insecurity percentage for each district.

The timing of publication creates another challenge. The 2024 CFSVA fieldwork took place between 26 April and 2 June 2024, but the final report was published in July 2025, approximately 14 months after the fieldwork began [3]. Although the results remain valuable, they describe conditions assessed more than a year before publication.

Shocks can also occur between survey rounds. Changes in rainfall, agricultural production and food prices may affect household food access before the next assessment is conducted. The 60% increase in food prices between April 2021 and May 2024 is one example of a major change occurring between the 2021 and 2024 survey rounds [3].

Meanwhile, several indicators associated with food security are monitored more frequently. NISR conducts seasonal agricultural surveys, rainfall is available through satellite-based datasets, temperature information is available through climate reanalysis, and WFP monitors food prices in local markets.

The challenge is therefore not simply the absence of data. It is the difficulty of integrating information from different sources and translating it into a useful estimate of household food insecurity at district level.

The CFSVA 2024 report itself recognises the importance of improving local monitoring. Its recommendations call for localised systems that monitor weather patterns and market conditions to help farmers and communities prepare for shocks such as droughts, floods and price increases [3].

HarvestWatch responds to this need by using existing information to estimate district-level food insecurity during years when a new household survey is unavailable.

1.4 Problem statement

Decision-makers in Rwanda need to plan agricultural and food-security interventions even when no recent representative household food-security measurement is available for a particular district.

Although information about rainfall, temperature, agricultural production and food prices is collected through different institutions and monitoring systems, these sources are not automatically translated into a comparable estimate of household food insecurity at district level.

As a result, districts experiencing worsening conditions may not receive additional attention until the next household survey or another assessment identifies the problem. This can reduce the time available for local investigation, preparedness and response.

The central problem is therefore the difficulty of combining existing information, identifying patterns associated with household food insecurity and presenting the resulting analysis in a form that decision-makers can understand and use.

1.5 Research question

The project is guided by the following research question:

Can historical CFSVA food-security data be combined with agricultural production, climate and market-price indicators to estimate the percentage of food-insecure households in each Rwandan district during the years between surveys, with sufficient accuracy to improve on a baseline that assumes the previous survey result remains unchanged?

The project also investigates whether interactive visualizations, model explanations and AI-assisted evidence retrieval can make these estimates more transparent and useful for decision-makers.

1.6 Proposed solution: HarvestWatch

HarvestWatch is a working early warning platform that combines machine learning, interactive data visualization and AI-assisted information retrieval.

The platform performs four connected functions.

First, it estimates the percentage of households experiencing moderate or severe food insecurity in each district for a selected year. The model produces a numerical percentage rather than directly predicting a risk category.

Second, the application converts the numerical estimate into a Low, Medium or High risk category. This makes it easier for users to compare districts while preserving the actual estimated percentage.

Estimated percentage of food-insecure households	**Risk category**	**Dashboard colour**
Below 15%	Low	Green
15% to below 30%	Medium	Yellow
30% or above	High	Red

These thresholds are project-defined communication categories. They are not official NISR, MINAGRI or IPC classifications and would require institutional validation before any formal operational adoption.

Third, HarvestWatch explains the estimates using SHAP. Users can investigate which model features contributed to a district's prediction and compare the model's historical estimates with genuine survey observations.

Finally, the platform includes an AI Analyst that retrieves relevant external evidence about weather, agricultural developments, food prices and other conditions. It helps users investigate a district's estimated risk, but it cannot change the numerical prediction generated by the Random Forest model.

Together, these functions turn different datasets into an accessible district-level monitoring tool.

1.7 Alignment with Rwanda's national development priorities

HarvestWatch is relevant to Rwanda's Second National Strategy for Transformation (NST2), covering 2024–2029. NST2 identifies priorities that include reducing stunting and malnutrition, improving agricultural productivity, strengthening climate resilience, advancing digital innovation and improving public service delivery [5], [6].

The strategy also targets agricultural growth of more than 6% annually and a 50% increase in the productivity of priority food crops, including maize, beans, potatoes, rice and cassava [6].

HarvestWatch supports these objectives by connecting agricultural production information with household food-security estimates. It tracks crop yields and cultivated areas, allowing users to investigate how agricultural conditions relate to the estimated food-security situation across districts.

The project also contributes to climate resilience by combining rainfall and temperature indicators with agricultural and market information. These features help users investigate whether unfavourable weather conditions may be associated with changing food-security estimates.

The Fifth Strategic Plan for Agriculture Transformation (PSTA5), covering 2024/25–2028/29, prioritises agricultural productivity, climate resilience, market integration and food security [8]. HarvestWatch supports these priorities by bringing together information about agricultural production, climate and staple-food prices in one platform.

The project is also relevant to Sustainable Development Goal 2, which focuses on ending hunger, achieving food security and improving nutrition. Its contribution is to support the availability and interpretation of information that may help institutions identify districts requiring closer monitoring.

National development priority	**HarvestWatch's contribution**
Reducing stunting and malnutrition	Identifies districts where estimated household food insecurity may be high, supporting further investigation.
Improving agricultural productivity	Combines crop yields, cultivated areas and agricultural input indicators with food-security estimates.
Strengthening climate resilience	Integrates rainfall and temperature indicators to help investigate potential climate-related pressures.
Digital innovation and public service delivery	Converts existing data into an accessible web-based decision-support platform.
Evidence-based planning	Provides numerical estimates, historical performance information and model explanations to support further investigation.

HarvestWatch is not a replacement for these national strategies or the institutions responsible for implementing them. It is a supporting tool designed to make existing information more useful for planning and monitoring.

2. Literature Review

2.1 Measuring household food security in Rwanda

The CFSVA uses the World Food Programme's Consolidated Approach for Reporting Indicators of Food Security (CARI) to classify households according to their food-security conditions.

CARI combines information about current food consumption with indicators of household coping capacity. These include the Food Consumption Score (FCS), food expenditure share, economic vulnerability and livelihood coping strategies.

Households are classified into four categories: food secure, marginally food secure, moderately food insecure and severely food insecure. The headline household food-insecurity percentage used by HarvestWatch is the sum of the moderately and severely food-insecure groups [3], [9].

The CFSVA is particularly important because it provides representative household food-security information at district level. However, its survey cycle limits the availability of new measurements between rounds.

The earlier survey rounds conducted in 2006, 2009 and 2012 used the Food Consumption Score rather than the full CARI classification. The FCS measures household food consumption by considering the frequency with which different food groups are consumed over a seven-day period [9].

HarvestWatch incorporates these earlier rounds through an estimated relationship between FCS-based observations and CARI. This extends the historical record available to the model, although the resulting estimates must be distinguished from direct CARI observations.

NISR's Seasonal Agricultural Survey provides another important source of information. It records agricultural production, cultivated area, yields and farming practices across Rwanda's districts and agricultural seasons [4]. However, agricultural production statistics alone do not measure whether households can afford or access sufficient food.

WFP's food-price monitoring provides information about the affordability of staple foods in local markets [10]. Similarly, CHIRPS rainfall and ERA5-Land temperature provide environmental indicators that help describe conditions affecting agricultural production [11], [12].

Regional early warning systems, including FEWS NET and the Integrated Food Security Phase Classification (IPC), provide information about acute food insecurity and crisis

conditions [13]. However, HarvestWatch has a different purpose: it uses Rwanda's historical CFSVA information to estimate a numerical household food-insecurity percentage for each of the country's 30 districts.

2.2 Machine learning for food-security prediction

Research has increasingly explored the use of secondary data, including satellite observations, market prices, agricultural statistics and household survey information, to estimate or forecast food-security conditions.

Andree and colleagues (2020) investigated the prediction of food crises using data from multiple countries. Their work demonstrated the potential of combining survey and other secondary information to identify food-security risks and provide policymakers with additional time to respond [14].

Foini and colleagues (2023) examined the forecastability of food insecurity using machine learning models and indicators related to food consumption, weather and economic conditions. Their findings highlighted that forecasting performance depends partly on the amount and quality of historical information available. They also demonstrated the importance of comparing machine learning models with simple baseline approaches rather than evaluating them in isolation [15].

Gholami and colleagues (2022) investigated food-security analysis and forecasting in Southern Malawi using machine learning. Their work provides a relevant example of how predictive models can be applied to food-security challenges in an African context [16].

These studies informed three important design decisions in HarvestWatch.

First, the project combines household food-security outcomes with secondary information about agricultural production, climate and market conditions.

Second, it evaluates the predictive model against a persistence baseline. This is important because food-security conditions may remain relatively stable in some districts, meaning that a complex model should demonstrate an improvement over simply carrying forward the previous survey result.

Third, the project uses a tree-based regression approach suitable for exploring relationships among different types of tabular data.

2.3 The gap HarvestWatch addresses

Existing food-security monitoring systems and research provide valuable information, but an applied tool is still needed to connect Rwanda's historical household food-security measurements with more frequently available agricultural, climate and market indicators.

Existing information or approach	Remaining gap
CFSVA provides representative district-level household food-security measurements every three years.	Additional district estimates are needed during the years between survey rounds.
NISR collects agricultural production and yield information through seasonal surveys.	Agricultural statistics need to be connected to household food-security outcomes.
CHIRPS and ERA5-Land provide climate information.	Climate indicators need to be interpreted alongside agricultural and market conditions.
WFP monitors staple-food prices.	Price changes need to be connected to their potential implications for household food access.
FEWS NET and IPC provide regional food-security assessments.	A complementary model based on Rwanda's own historical survey data can provide estimates for individual districts.
International research demonstrates the potential of machine learning for food-security forecasting.	Rwanda needs an applied, evaluated and accessible tool built around its own data and district structure.

HarvestWatch addresses this gap by integrating existing datasets into a working platform that produces district-level estimates, explains model behaviour and provides historical performance information.

Its contribution is not simply the use of a machine learning algorithm. It is the integration of national and international data sources into an accessible workflow that supports investigation between official survey rounds.

3. Data and Methodology

3.1 Data sources

HarvestWatch combines five main data sources. Each source contributes information about a different dimension of household food security.

Data source	Provider	Variables used	Coverage in assembled data
Comprehensive Food Security and Vulnerability Analysis (CFSVA)	NISR, MINAGRI and WFP	Percentage of households experiencing moderate or severe food insecurity; historical FCS information	2006–2024
Seasonal Agricultural Survey (SAS)	NISR	Crop yields, cultivated areas, agricultural production, improved seed, fertilizer use and share of harvest sold	2013–2025
CHIRPS rainfall	Climate Hazards Group, through WFP/HDX	Annual and seasonal rainfall, rainfall anomalies, dry and very wet month indicators	1981–2026
ERA5-Land temperature	Copernicus/ECMWF	Average and seasonal temperature indicators	1995–2026
WFP food-price data	WFP, through Humanitarian Data Exchange	Retail prices of beans, maize, cassava flour, Irish potatoes and sorghum	2008–2026

CFSVA household data

The CFSVA provides the historical target variable used by the model. The project works with household-level survey information to calculate the survey-weighted percentage of households classified as moderately or severely food insecure in each district.

The 2015, 2018, 2021 and 2024 rounds provide CARI-based observations. The earlier rounds from 2006, 2009 and 2012 provide FCS information, which is converted into estimated CARI-equivalent values for use in the extended historical dataset.

Seasonal Agricultural Survey

The SAS provides agricultural production indicators for crops including maize, beans, cassava, Irish potatoes, sweet potatoes, sorghum, bananas and rice.

The project uses variables such as cultivated area, crop yield, agricultural production, improved seed use, fertilizer use and the share of harvest sold. These indicators help describe agricultural conditions that may influence household food availability and income.

CHIRPS rainfall

CHIRPS provides rainfall estimates by combining satellite observations with station data [11]. HarvestWatch uses this information to construct district-level rainfall indicators, including annual and seasonal rainfall, rainfall anomalies and indicators of unusually dry or wet months.

ERA5-Land temperature

ERA5-Land provides climate reanalysis information for land applications [12]. HarvestWatch uses its temperature information to derive indicators such as average temperature, seasonal temperature and unusually warm periods.

WFP food prices

The project uses WFP market-price information for staple foods including beans, maize, cassava flour, Irish potatoes and sorghum [10]. These variables help represent changes in the cost of food and the potential pressure that rising prices may place on household purchasing power.

Some price observations are recorded at market or regional level rather than independently for every district. They are therefore interpreted as indicators of broader market conditions, not exact measurements of the prices paid by every household in a district.

3.2 Dataset construction and integration

The data pipeline combines the five sources into a single district-year modelling table.

The processing workflow is organised into six steps under the project's `pipeline/` directory and is coordinated through `pipeline/run_all.py`. The final output is `food_security_dataset.csv`.

The resulting dataset contains 630 records covering 30 districts over 21 years, from 2006 to 2026. Each row represents one district in one year.

Dataset component	**Number of records**	**Purpose**
Historical district-year records, 2006–2024	570	Model training and evaluation
Forecast district-year records, 2025–2026	60	Generate estimates for years without CFSVA observations
Total	**630**	Complete modelling dataset

The 45 model features include year, province, district, historical food-security information, rainfall indicators, temperature indicators, agricultural indicators and staple-food prices.

The sources originally use different geographical structures and naming conventions. The pipeline standardises district and province names before combining the information. Records are aligned using available geographic identifiers, including year, province, district and district code.

The pipeline also addresses differences in temporal coverage. For example, WFP price data begins in 2008, while the assembled SAS data begins in 2013. Climate datasets cover longer historical periods.

These differences mean that not every feature is available for every district-year record. The modelling pipeline therefore includes missing-value handling and preprocessing.

3.3 Forecasting window and temporal alignment

An important methodological decision is ensuring that the model uses information that would be available before the period being estimated.

The project aligns each prediction year to the approximate April–May CFSVA fieldwork period. It uses a 12-month feature window covering April of the previous year through March of the selected year.

For example, when estimating food insecurity for 2026, the model uses relevant information from April 2025 through March 2026.

This window covers agricultural, climate and market conditions leading up to the assessment period. It is designed to reduce the risk of using information collected after the period the model is intended to estimate.

Temporal alignment is particularly important in forecasting. If a model uses future information that would not have been available at the time of prediction, its evaluation may appear stronger than its real-world performance.

3.4 Target construction and data provenance

The model's target variable is the percentage of households in a district experiencing moderate or severe food insecurity.

However, the historical survey record does not contain a direct CARI observation for every district in every year. The project therefore constructs the historical target through three processes.

For the 2015, 2018, 2021 and 2024 CFSVA rounds, the project calculates the survey-weighted percentage of households classified as moderately or severely food insecure using CARI.

For the earlier rounds in 2006, 2009 and 2012, the project uses the weighted share of households with poor or borderline Food Consumption Scores and converts these values into CARI-equivalent estimates using a linear relationship estimated from the later CARI observations.

For years between survey rounds, the pipeline uses linear interpolation within districts. Values before a district's first available survey observation are back-filled.

The resulting historical target distribution is:

Target source	**Number of records**	**Use**
Direct CARI survey observations	120	Model training and genuine-survey evaluation
FCS-based survey estimates	86	Model training

Interpolated values	358	Model training only
Back-filled values	6	Model training only
Total historical records	**570**	

This distinction is essential when interpreting the model's performance. Although the historical dataset contains 570 labelled district-year records, only 120 are direct CARI survey observations. The remaining 450 values are derived through FCS-based estimation, interpolation or back-filling.

The model is therefore not evaluated by treating every derived value as independent ground truth. Its headline evaluation uses genuine CARI survey observations from 2021 and 2024.

Each historical record also contains a `label_source` field that identifies whether the target is a direct CARI observation, an FCS-based estimate, an interpolated value or a back-filled value. This information supports transparency within the application.

3.5 Data preprocessing

Before model training, the pipeline applies data-quality checks and preprocessing.

Geographical identifiers are standardised to ensure that the same district is represented consistently across different sources. Duplicate records, inconsistent units and invalid values are checked during data preparation. Infinite values resulting from calculations such as yield changes after a zero-yield observation are converted into missing values.

The production preprocessing pipeline uses median imputation for numeric features, scaling and one-hot encoding for categorical features such as district and province.

These transformations are applied consistently when preparing model inputs for prediction.

Missing values are expected because the data sources have different coverage periods and reporting frequencies. The application also indicates when forecast inputs are incomplete or have been imputed.

This is particularly relevant for 2026 because Season A 2026 agricultural survey data was not available when the forecasts were prepared. The relevant agricultural inputs therefore rely on available Season B and Season C 2025 information.

3.6 Machine learning model: Random Forest

HarvestWatch uses a Random Forest implemented with scikit-learn.

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. Each tree learns relationships between the input features and the target variable, and the model combines their predictions to produce an overall estimate.

The approach is useful for investigating food-security relationships because agricultural, climate and market indicators may interact in non-linear ways. For example, the relationship between rainfall and agricultural production may depend on the timing of rainfall, temperature conditions, crop type and previous agricultural conditions.

Random Forest can capture such patterns without requiring the team to manually specify every possible relationship.

The selected model configuration is:

Parameter	**Value**
Model	RandomForest
Number of trees	500
Maximum tree depth	7
Minimum samples per leaf	2
Maximum features considered	`sqrt`
Split criterion	`absolute_error`
Random state	42

The model receives the prepared district-year features and estimates a continuous percentage of food-insecure households.

The application subsequently applies the project's risk thresholds to this numerical prediction. The Random Forest model is therefore performing regression, not directly predicting one of three categories.

This approach preserves the numerical estimate while making it easier for users to compare districts through the risk bands.

3.7 Model training and evaluation

HarvestWatch is intended to estimate conditions in years that were not covered by a new household survey. Its evaluation must therefore test how well it performs on later observations that were not used to train the evaluation model.

The project uses a time-based evaluation strategy rather than relying only on a random train-test split. This is important because observations from the same district in nearby years may be similar. A random split could allow information from later periods to influence training and make the model appear more accurate than it would be when predicting genuinely later years.

The evaluation model was trained on the 450 historical district-year records from 2006 through 2020. It was then evaluated against 60 genuine CARI observations from the 2021 and 2024 CFSVA rounds, with 30 district observations in each round.

The project also compares the model with a persistence baseline. This baseline assumes that the most recent available survey result remains unchanged. It represents a simple alternative to using the machine learning model and helps establish whether the additional data and modelling provide practical predictive value.

After evaluation, the final model was trained using the complete historical dataset from 2006 through 2024, containing 570 labelled records. It was then used to generate estimates for 2025 and 2026.

The evaluation metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2.

MAE measures the average absolute difference between predicted and observed percentages. RMSE gives greater weight to larger errors, making it useful for identifying substantial overestimation or underestimation. R^2 indicates how much variation in the observed values is explained by the model relative to a mean-based reference.

Overall evaluation results

Evaluation set	Model	MAE (percentage points)	RMSE (percentage points)	R^2

2021 and 2024 CARI observations	Random Forest	5.39	6.77	0.62
2021 and 2024 CARI observations	Persistence baseline	7.32	9.29	0.28
2021 CARI observations	Random Forest	6.33	7.71	0.64
2021 CARI observations	Persistence baseline	7.36	8.95	0.51
2024 CARI observations	Random Forest	4.44	5.68	0.55
2024 CARI observations	Persistence baseline	7.29	9.61	-0.28

Across the 60 genuine survey observations, the Random Forest model achieved an MAE of 5.39 percentage points, compared with 7.32 percentage points for the persistence baseline. This represents approximately 26% lower average absolute error.

The improvement was particularly visible in the 2024 evaluation. The model achieved an MAE of 4.44 percentage points, compared with 7.29 for the baseline, representing approximately 39% lower error.

The model also produced an estimate within five percentage points of the observed survey result in approximately 57% of the evaluated cases, compared with 43% for the persistence baseline.

These results suggest that combining historical household food-security information with agricultural, climate and market indicators provides additional predictive value compared with simply carrying forward the previous survey result.

Risk Classification Performance

In addition to evaluating the model's numerical predictions, we assessed how effectively the system translates predicted food-insecurity percentages into the three project-defined risk categories: Low, Medium and High.

This evaluation is important because district officials and other stakeholders may use these categories to identify areas requiring closer monitoring. While regression metrics measure how close the predicted percentages are to the observed survey values, classification metrics show how reliably the system assigns districts to the appropriate risk categories.

Overall risk-classification performance

Metric	**Test data 2021–2024**	**Real surveys 2021 & 2024**	**Real survey 2021**	**Real survey 2024**
Accuracy	70.0%	66.7%	60.0%	73.3%
Precision	72.7%	63.1%	74.6%	51.7%
Recall	61.9%	55.1%	55.4%	51.7%
F1-score	64.9%	57.4%	58.2%	51.7%
ROC-AUC	89.2%	84.7%	85.6%	87.6%

The overall results show that the model achieved 70.0% accuracy on the 2021–2024 test data. When evaluated specifically against the 60 genuine CARI observations from the 2021 and 2024 surveys, accuracy was 66.7%, with a ROC-AUC of 84.7%.

The genuine-survey results are particularly important because they measure performance against actual survey observations rather than relying on the full historical dataset, which also contains estimated and interpolated target values.

Class-level performance on genuine 2021 survey observations

Risk class	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Low	71.4%	41.7%	52.6%	87.0%

Medium	52.4%	84.6%	64.7%	73.8%
High	100.0%	40.0%	57.1%	96.0%
Average	74.6%	55.4%	58.2%	85.6%

The 2021 results show that the model achieved 100.0% precision for the High-risk category, meaning that its High-risk predictions were correct. However, recall was 40.0%, indicating that it identified only a proportion of all genuinely High-risk observations. The Medium category had the highest recall at 84.6%.

Class-level performance on genuine 2024 survey observations

Risk class	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Low	75.0%	75.0%	75.0%	93.2%
Medium	80.0%	80.0%	80.0%	80.2%
High	0.0%	0.0%	0.0%	89.3%
Average	51.7%	51.7%	51.7%	87.6%

The 2024 results show stronger performance in the Low and Medium categories. However, the model did not correctly identify any of the genuine High-risk observations at the selected classification threshold. This is a significant limitation because failing to identify a High-risk district could lead to an area requiring attention being assigned a lower risk category.

Although the High-risk ROC-AUC was 89.3%, this does not mean that the model correctly classified High-risk observations at the selected threshold. ROC-AUC measures the model's ability to distinguish between classes across different thresholds, whereas precision, recall and F1-score measure performance at the threshold actually used.

Overall, these results demonstrate that HarvestWatch can provide useful district-level risk screening, but its classification performance is not equally strong across all risk categories.

In particular, the identification of High-risk districts requires further improvement. The risk categories should therefore support prioritisation and further investigation rather than replace official food-security assessments or field verification.

Evaluation limitations and interpretation

The evaluation results should be interpreted alongside the limitations of the historical target construction.

The 2019 and 2020 training targets were interpolated using the 2021 CFSVA observations. Consequently, the 2021 evaluation may be somewhat optimistic because the survey observations being evaluated contributed indirectly to the construction of some training targets.

The 2024 evaluation is therefore particularly important when interpreting the model's performance on a later survey round.

The model also does not perform equally well across all risk levels. Of the seven genuine High-risk observations in the evaluation set, only two were classified as High by the model. The remaining five were placed in Medium. This indicates that the model can underestimate the most severe food-insecurity conditions.

The results demonstrate improvement over the persistence baseline, but they do not establish that the model will perform equally well in every future year. Continued evaluation against new genuine CARI observations will be necessary.

4. Forecast Results

4.1 2025 forecasts

The final model generated estimates for all 30 districts for 2025. The average district estimate was approximately 18.9%.

Under the project's risk thresholds, seven districts were classified as Low, 22 as Medium and one as High.

The highest 2025 estimates included:

District	**Estimated food-insecure household share**
Ngororero	31.0%

Rutsiro	28.6%
Rubavu	28.4%
Nyamasheke	27.3%

These figures are model-generated estimates rather than official CFSVA observations. They are intended to provide an additional signal for identifying districts that require closer investigation.

4.2 2026 forecasts

For 2026, the model estimated an average district food-insecurity percentage of 19.6%.

Under the project's thresholds, eight districts were classified as Low, 19 as Medium and three as High.

The three districts with High estimates were:

District	**Estimated food-insecure household share**	**Risk category**
Rutsiro	32.7%	High
Ngororero	31.8%	High
Nyamasheke	30.0%	High

Nyamagabe had an estimate of 29.6%, placing it close to the High threshold.

The estimates also indicate that 17 of the 30 districts are projected to have higher food-insecurity percentages than their 2024 survey values. This provides another way to identify districts where conditions may warrant closer attention, even when their estimates remain within the Medium category.

These figures should be interpreted as model outputs rather than official measurements. In particular, the 2026 agricultural inputs are incomplete because Season A 2026 SAS data was not yet available when the forecasts were prepared.

4.3 Interpreting the forecast results

The forecasts provide an additional way for decision-makers to investigate district-level conditions between CFSVA rounds.

For example, Rutsiro, Ngororero and Nyamasheke appear in the High category for 2026 under the project's thresholds. A district planner could examine the numerical estimate, review the leading SHAP contributions, compare rainfall and agricultural production trends, and investigate food-price changes.

The AI Analyst can then retrieve external evidence that may help explain recent developments relevant to the district.

This process does not establish that the forecast is the actual food-insecurity rate. It provides a starting point for further investigation, consultation and field verification.

The numerical estimate should always be considered alongside its historical performance, input completeness and the limitations of the model.

5. Technology Innovation and Implementation

5.1 System architecture

HarvestWatch combines data processing, live data visualization, machine learning, fullstack web engineering, geospatial visualization and AI-assisted information retrieval in one application.

The system follows a pipeline in which source datasets are processed into a district-year modelling table. The trained model and its preprocessing components are saved as model artifacts. The backend loads the prepared data and saved model when the application starts, and the frontend communicates with the backend to retrieve forecasts, historical results and explanations.

The architecture separates the numerical prediction from the user interface and AI-assisted interpretation. This allows the model to generate estimates independently of the language model used by the AI Analyst.

Component	Technology	Implemented responsibility
Data pipeline	Python	Processes and combines CFSVA, SAS, rainfall, temperature and food-price data.
Frontend	React, TypeScript, React Router, Vite and Lucide	Provides maps, dashboards, filters, tables, charts and interactive forms.
Backend	FastAPI and Uvicorn	Serves data and predictions, validates inputs and connects application services.
Machine learning	scikit-learn Random Forest pipeline	Generates historical estimates, forecasts and scenario predictions.
Explainability	SHAP TreeExplainer	Provides global feature importance and district-level prediction contributions.
Geospatial visualization	Rwanda district GeoJSON	Supports district-level mapping and risk visualization.
AI agent	Google Agent Development Kit (ADK)	Organises the AI Analyst's interpretation and research workflow.
Language-model integration	LiteLLM and OpenRouter	Provides access to the configured language model.
External research	Tavily Search API	Retrieves relevant external evidence and source links.

Deployment	Docker and AWS	Packages and deploys the frontend and backend as one web service.
Testing	pytest	Tests API behaviour, model results, SHAP explanations, validation and AI safeguards.

The current application uses local datasets and saved model artifacts rather than a separate database service. The backend loads the prepared data and models at startup and precomputes historical predictions, forecasts and SHAP values. It does not retrain the model during individual user requests.

5.2 Prediction service and validation

The FastAPI backend provides the services required by the frontend, including model status, geographical information, national overview data, historical results, model performance, feature importance, forecasts, district time series, visualizations, Prediction Lab requests and AI Analyst requests.

When a user requests a prediction, the backend prepares the district's input features, applies the same preprocessing steps used during training and sends the prepared information to the Random Forest model.

The model returns a numerical percentage, which the application converts into the corresponding risk category.

The backend also validates user inputs. Unknown districts, unsupported years, unknown features and invalid negative values are rejected. When inputs are missing, the preprocessing pipeline fills them using the configured imputation process, and the response identifies which values were imputed.

This helps prevent invalid information from being passed into the model without users being informed.

5.3 Prediction Lab: interactive scenario testing

The Prediction Lab allows users to explore how changes in selected input variables affect the model's estimate.

A user selects a district and year, reviews the model's default inputs and changes selected variables. For example, the user may reduce rainfall or increase the price of maize to examine how the model responds under a hypothetical scenario.

The backend validates the modified inputs, applies the model's preprocessing steps and generates a new estimate. The application then displays the updated percentage, risk category and explanation.

This feature makes the model more interactive and allows users to investigate hypothetical agricultural and market conditions without needing to work directly with the underlying dataset or machine learning code.

However, a scenario result is not a causal prediction. Changing rainfall or food prices in the Prediction Lab shows how the trained model responds to modified inputs; it does not prove what would happen in the real world if those conditions occurred.

5.4 Explainable machine learning with SHAP

HarvestWatch integrates SHAP using TreeExplainer to make the Random Forest model easier to investigate.

The application presents global feature importance and district-level contributions to individual predictions. These explanations help users identify which features the model relied on when producing an estimate.

For example, users can investigate whether historical food-security information, rainfall, temperature, agricultural production or market prices contributed to a district's prediction.

This is important because a food-security estimate should not be presented as an unexplained number. Users need to understand which information influenced the result and whether the estimate may be affected by incomplete or imputed inputs.

SHAP explanations describe model behaviour, not causation. A feature that contributes strongly to a prediction is not automatically the cause of food insecurity. The explanations are therefore intended to support interpretation and investigation rather than establish causal relationships.

5.5 AI Analyst and evidence retrieval

The AI Analyst is an additional feature designed to help users investigate the context surrounding a district's estimate.

It is implemented using a Google ADK language-model agent, with model access through LiteLLM and OpenRouter. It uses the Tavily Search API to retrieve relevant external information.

For a selected district and year, the backend provides the agent with fixed model context. This includes the numerical prediction, risk category, leading SHAP contributions, historical survey information and relevant climate, agricultural and food-price indicators.

The AI Analyst uses this information to produce a structured interpretation covering the estimate, contributing factors, relevant external evidence, conditional implications, possible recommendations and limitations.

Because language models can produce convincing but unsupported statements, the system includes several safeguards.

The most important safeguard is that the AI Analyst cannot change the numerical prediction. The Random Forest model generates the estimate, and the AI agent receives it as fixed information.

The system also verifies citations against the external search results retrieved during the analysis. Sources are assigned identifiers, and citations that do not match retrieved results are removed before the answer is displayed.

Statements about future conditions are presented as conditional scenarios rather than validated forecasts. For example, the AI Analyst may discuss what a future year could look like under certain conditions, but it does not present that scenario as an official prediction.

To control external API costs, identical requests can be served from a cache, the number of searches per analysis is limited, and hourly and daily usage limits are applied. If required external services are unavailable, the application reports the problem rather than presenting unsupported information.

The AI Analyst therefore provides an additional interpretation and research layer. It does not replace the numerical model or independently validate its predictions.

5.6 Testing and reliability

The project includes 15 automated tests implemented with pytest, and all 15 pass in the documented test run.

The tests cover the backend's health and status responses, model performance calculations, the availability of forecasts for all 30 districts in 2025 and 2026, and the consistency of SHAP explanations with the corresponding model predictions.

Additional tests check input validation in the Prediction Lab, the correctness of visualization summaries, usage limits, AI citation filtering and the serving of the completed web application.

The testing process is important because the figures presented in the report should remain consistent with those displayed in the application. Tests that verify model metrics and prediction outputs help reduce the risk of reporting results that do not match the saved model artifacts.

The tests do not establish that every possible real-world situation has been covered. However, they provide a repeatable way to check important application functions and safeguards as the project develops.

5.7 What makes HarvestWatch innovative

The main innovation in HarvestWatch is the way it integrates existing information into a working district-level early warning platform.

The first contribution is filling the information gap between household surveys. Instead of waiting for another CFSVA round, the platform uses available agricultural, climate and market indicators to estimate district-level food insecurity during the intervening years.

The second contribution is connecting seven historical survey rounds. By incorporating earlier FCS-based information through an estimated relationship with CARI, the project extends the available historical record back to 2006 while preserving labels that distinguish direct observations from derived estimates.

The third contribution is data transparency. Historical values are labelled according to their source, and the application distinguishes survey observations, survey estimates, interpolated values, back-filled values and forecasts. This allows users to understand where each number comes from.

The fourth contribution is explainability. stakeholders can see the numerical estimate, its historical performance and the features contributing to individual predictions rather than relying only on a risk category.

Finally, HarvestWatch combines machine learning with AI-assisted research while keeping the two functions separate. The Random Forest model provides the numerical estimate, while the AI Analyst retrieves external evidence and helps interpret the result without being able to override the model.

6. Usability and Design

HarvestWatch is designed for decision-makers, planners, analysts and development partners who may not have a machine learning background. The application presents the results through maps, numerical estimates, charts and explanations rather than requiring users to work directly with the underlying datasets.

The interface is organised around the questions that a user may need to answer.

The **Overview** page provides a national picture for a selected year. It displays the number of districts in each risk category, the districts with the highest estimates, trends and a map of Rwanda's 30 districts. Users can filter the information by province or district.

The **Forecasts** page presents the 2025 and 2026 estimates for all 30 districts. It displays each numerical percentage, its risk category, the 2024 observed value for comparison, leading SHAP contributions and information about the completeness of the forecast inputs. It also makes clear that these years do not have new CFSVA observations.

The **Historical** page allows users to compare model estimates with available survey observations from 2021 and 2024. It displays district-level errors and allows users to investigate the model's performance rather than relying only on an overall accuracy score.

The **Prediction Lab** allows users to change selected input variables and examine how the model responds to hypothetical scenarios. This makes the model more accessible to users who want to investigate possible changes in rainfall, prices or agricultural conditions.

The **Visualizations** page provides interactive exploration of rainfall, temperature, food prices, crop production and yields across selected years, districts and provinces. It helps users investigate the conditions that may be associated with changes in food-security estimates.

The **AI Analyst** provides source-linked explanations and external evidence to help users investigate the model's estimates further.

Several design principles guide the application. Risk categories are never communicated through colour alone; the words Low, Medium and High are displayed alongside the percentage. Survey observations and model-generated estimates are visually distinguished. Charts and tables identify their units, time periods and geographical levels. Limitations are also communicated where they matter, including the illustrative nature of the risk thresholds and the incomplete agricultural inputs for 2026.

These design choices aim to make the platform understandable without hiding the uncertainty behind its predictions.

7. Tangible Impact and Stakeholder Value

7.1 Intended beneficiaries

HarvestWatch is designed to support institutions involved in agricultural planning, food-security monitoring, social protection and humanitarian preparedness.

For **MINAGRI**, the platform can help identify districts with high or rising food-insecurity estimates and provide information about the agricultural and climate indicators associated with those estimates. This may support further investigation into crop production, agricultural inputs, irrigation and post-harvest conditions.

For **NISR**, HarvestWatch provides an additional way to explore district-level conditions during years without a new CFSVA observation. Its historical comparisons and data-source labels may help identify districts where conditions appear to have changed and where further statistical investigation could be useful.

For **district authorities**, the platform provides district-specific estimates and visualizations that can support local monitoring, discussions about emerging risks and requests for further assessment.

For **WFP and humanitarian partners**, the platform can provide an additional signal for identifying potential hotspots and investigating the relationship between food prices, climate conditions and household food insecurity.

For **MINALOC and social-protection stakeholders**, district estimates may help identify areas where households could be vulnerable to moving from marginal food security into food insecurity, supporting further investigation and preparedness.

For **researchers and development partners**, the documented district-year dataset and model explanations provide an opportunity to investigate relationships between climate, agricultural production, food prices and household food security.

7.2 Practical example: using the 2026 forecast

The 2026 forecast identifies Rutsiro, Ngororero and Nyamasheke as High under the project's thresholds. Nyamasheke was also among the districts with the highest food-insecurity levels in the 2024 CFSVA [3].

Without an additional monitoring tool, decision-makers would need to wait for another official household survey to obtain a new representative measurement for these districts. With HarvestWatch, a planner can examine the 2026 estimate, compare it with the 2024 survey result and investigate the factors contributing to the model's prediction.

The planner can then use the Visualizations page to review rainfall, agricultural production and food-price trends. The AI Analyst can retrieve relevant external evidence to support further investigation.

The same workflow can help identify districts that are not yet classified as High but have experienced an increase in their estimated food-insecurity percentage. This is important because a district does not necessarily need to cross the High threshold before its conditions deserve attention.

The appropriate next step is to verify the information through consultation, local knowledge or field assessment. The forecast is a signal for investigation, not an automatic instruction to allocate resources.

7.3 Expected impact

The most direct potential contribution of HarvestWatch is earlier attention to districts that may be experiencing worsening food-security conditions.

The model provides a numerical estimate for every district during years when a new CFSVA result is unavailable. Its average error was approximately 26% lower than the persistence baseline across the 60 genuine survey observations. This suggests that the combined data and modelling approach can provide a more informative estimate than simply assuming that the previous survey result remains unchanged.

The platform also supports more transparent investigation. Users can examine the numerical estimate, its historical accuracy, the source of the target information and the features contributing to the prediction. This makes it easier to understand the basis of an estimate and recognise where caution is necessary.

Another potential contribution is making better use of existing data. NISR, MINAGRI, WFP and other providers already collect information about household food security, agricultural production, rainfall, temperature and food prices. HarvestWatch creates additional value by integrating these sources into one analytical platform without requiring a new household survey for every prediction year.

The project is also designed to be extendable. When new CFSVA results become available, the historical dataset can be updated, the model retrained and its performance evaluated against the new genuine observations.

However, these are potential contributions rather than measured social outcomes. HarvestWatch has not itself delivered food assistance, agricultural inputs or cash transfers, and the project has not established that its forecasts have already reduced food insecurity. Its impact will depend on whether relevant institutions adopt the information, verify emerging risks and take appropriate action.

A practical next step towards real-world use would be a pilot with selected district authorities and relevant national institutions. The pilot could compare the platform's estimates with field observations and determine how the information could complement existing monitoring processes.

8. Limitations and Responsible Use

8.1 Limitations

HarvestWatch demonstrates the potential of combining historical food-security information with agricultural, climate and market indicators. However, several limitations affect the reliability and interpretation of its estimates.

Limited genuine survey observations. The historical dataset contains 570 labelled records, but only 120 are direct CARI observations. A further 86 are FCS-based estimates, while 358 are interpolated and six are back-filled. The model therefore learns from a dataset containing a substantial number of derived target values. Its evaluation is based on 60 genuine CARI observations from 2021 and 2024.

Evaluation leakage risk. The 2019 and 2020 training targets were interpolated using the 2021 CFSVA observations. This means that the 2021 evaluation may be somewhat optimistic. The 2024 evaluation provides an important additional indication of performance on a later survey round.

Underestimation of High-risk districts. The model classified only two of seven genuine High-risk observations as High, placing the other five in Medium. This shows that the model can underestimate the most severe food-insecurity conditions. Users should therefore examine the numerical estimates and historical performance rather than relying only on the risk category.

Dependence on historical geographic patterns. District and province identity, together with previous food-security information, contribute substantially to the model's predictions. This indicates that the model captures persistent geographical patterns as well as relationships involving climate, agriculture and market conditions. It may therefore be less responsive to sudden changes that differ significantly from historical patterns.

Different geographical coverage. Some food-price and agricultural indicators are available at market or regional level rather than independently for every district. These variables provide useful context but cannot fully represent differences between individual households or markets within a district.

Incomplete agricultural inputs for 2026. Season A 2026 SAS data was not available when the forecasts were prepared. Relevant agricultural features therefore rely on the available Season B and Season C 2025 information.

Project-defined risk thresholds. The Low, Medium and High categories are communication thresholds selected by the project. They are not official government classifications and require validation with relevant institutions before any formal operational use.

Unprecedented shocks. The model learns from historical relationships and may not fully anticipate events that have no meaningful representation in its training data. Major environmental, economic or other shocks could cause actual conditions to differ from the estimates.

External AI dependencies. The AI Analyst depends on external services, including OpenRouter and Tavily. Its availability may be affected by service interruptions, missing API credentials, usage limits and the quality of external information. Retrieved evidence provides context but does not independently validate a model prediction.

8.2 Responsible use

Food-security estimates can influence decisions involving vulnerable households and limited public resources. An incorrect estimate could direct attention away from a district that needs assistance or create an inaccurate impression of local conditions.

HarvestWatch is therefore designed as a decision-support tool rather than an automatic resource-allocation system.

The application displays historical model performance and known weaknesses. Risk categories are labelled as project-defined rather than official classifications. Survey

observations, derived estimates and forecasts are distinguished so that users can understand the source of each value.

SHAP explanations are presented as descriptions of model behaviour, not proof of causation. AI-generated interpretations include source links and cannot alter the numerical prediction. Future scenarios are presented as conditional possibilities rather than official forecasts.

Most importantly, the platform does not recommend that assistance be allocated solely on the basis of its predictions. Official CFSVA results, field assessments, administrative information and local knowledge remain essential for final decisions.

8.3 Future improvements

Several improvements could strengthen HarvestWatch's reliability and practical value.

The first is to retrain and evaluate the model when the 2027 CFSVA becomes available. New genuine survey observations would provide an opportunity to assess whether the model continues to improve on the persistence baseline.

The second is to improve the identification of the most severely affected districts. Future work could investigate alternative model configurations, training strategies and additional indicators that better capture sudden increases in food insecurity. The platform could also provide a watch list of districts whose estimates or risk probabilities indicate a need for closer investigation.

The third is to introduce uncertainty ranges. Instead of displaying only one percentage, future versions could provide a range of plausible estimates so that users can understand how uncertain a prediction may be.

The fourth is to strengthen evaluation. The team could reconstruct the 2019–2020 training targets without using the 2021 survey and conduct additional time-based validation across available survey rounds. This would help determine how much the current evaluation is affected by the target-construction process.

The fifth is to improve the geographical and temporal detail of the input data. More detailed market-price information, livestock indicators and additional agricultural statistics could improve the representation of local conditions. Season A 2026 SAS information should also be incorporated when it becomes available.

Finally, the project should work with NISR, MINAGRI and relevant district institutions to validate the risk thresholds and determine how HarvestWatch could complement existing food-security monitoring systems. A pilot with selected district authorities would be a practical way to compare the model's warnings with field observations before any wider adoption.

9. Conclusion

Rwanda collects valuable information about household food security, agricultural production, climate conditions and food prices. However, these sources are produced at different frequencies and for different purposes. The CFSVA provides representative household food-security measurements, but its approximately three-year survey cycle creates periods when decision-makers do not have a new district-level measurement, even as agricultural and market conditions continue to change.

HarvestWatch addresses this information gap by integrating historical CFSVA information with NISR agricultural statistics, CHIRPS rainfall, ERA5-Land temperature and WFP food prices. The resulting dataset contains 630 district-year records covering all 30 districts from 2006 to 2026, with 45 features used by the predictive model.

The Random Forest estimates the percentage of households experiencing moderate or severe food insecurity. The application then converts the numerical estimate into a project-defined risk category and presents the result alongside historical information and model explanations.

The evaluation model achieved an MAE of 5.39 percentage points across 60 genuine CARI observations from 2021 and 2024, compared with 7.32 percentage points for the persistence baseline. This represents approximately 26% lower average error. The 2024 evaluation showed an MAE of 4.44 percentage points compared with 7.29 for the baseline. These results demonstrate that the combined approach provides additional predictive value, while the model's tendency to underestimate some High-risk districts highlights the importance of transparency and continued evaluation.

The completed React and FastAPI platform makes these estimates accessible through national and district-level dashboards, forecasts, historical comparisons, interactive visualizations, a Prediction Lab, SHAP explanations and an AI Analyst that retrieves source-linked evidence. Its automated tests provide additional checks on the reliability of important application functions.

HarvestWatch's main contribution is turning information that Rwanda already collects into a more connected and accessible district-level early warning resource. It supports national priorities related to agricultural productivity, climate resilience, food security and evidence-based public planning.

The project does not claim to replace official surveys or field assessments, nor does it claim that a forecast is equivalent to a measured household food-insecurity rate. Instead, it provides an additional signal that can help institutions identify where conditions may be changing, investigate the available evidence and determine whether further assessment is needed.

Ultimately, HarvestWatch aims to reduce the information gap between household surveys by helping decision-makers notice potential food-security concerns earlier and make better-informed decisions using existing data.

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